

Rishi More

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Summary

ML engineer focused on reliable and efficient **LLM systems**, with hands-on experience in **CUDA kernel profiling**, **vLLM**-based calibration/evaluation pipelines, temporal RAG, and low-latency ML deployment. **Co-first author of an ICML 2026 paper** on distribution-free risk control for reducing test-time compute under calibrated error guarantees. Experienced in turning research prototypes into production-facing ML pipelines and evaluation harnesses.

Experience

JHU Data Science and AI Institute (DSAI)

May 2025 – Present

Graduate Research Assistant, LLM Inference Optimization

Baltimore, MD

- **Co-first author** of *Conformal Thinking (ICML 2026)*: distribution-free risk control setting dual confidence thresholds for adaptive reasoning, lowering inference cost while bounding error at a user-specified risk level.
- Built a vLLM-based calibration/evaluation pipeline across **4** reasoning models and **4** benchmarks (**2.7K+** problems), maintained target error rates under dataset/length shift using finite-sample (UCB) calibration.
- Designed dual-threshold stopping rules, including a parametric lower threshold and ensembled halt policy, reducing reasoning tokens by **12%** at matched accuracy with reusable evaluation tooling.

JHU Center for Language and Speech Processing (CLSP)

January 2025 – December 2025

Graduate Researcher, Retrieval Systems

Baltimore, MD

- Implemented a temporal-aware RAG pipeline over **10K+** non-stationary documents using time-decay weighting, metadata-aware indexing, and temporally scoped retrieval.
- Improved retrieval precision by **23%** over dense baselines in high-volatility knowledge domains by prioritizing time-relevant evidence.
- Reduced false retrievals by **31%** using reference-free consistency scoring to detect stale or hallucination-prone retrieved context.

Mastek

November 2022 – July 2023

Machine Learning Engineer Intern

Mumbai, India

- Built and deployed an on-prem **edge** voice authentication system at **<100ms p99** latency and **96.6%** accuracy, using a lightweight Siamese mel-spectrogram model for retraining-free enrollment under edge-compute limits.
- Engineered a TensorFlow pipeline over **1M+** voice samples with spectral-gating denoising for robust feature extraction.
- Improved authentication reliability **20%** in high-noise production environments via preprocessing and feature-extraction upgrades.

Projects

Post-Training Evaluation & Reward-Hacking Detector | Python, PyTorch, DPO, LLM-as-Judge, Calibration | GitHub

- Built an evaluation harness for **6** post-training failure modes—reward hacking, sycophancy, verbosity bias, false confidence, safety drift, and distribution shift across **4** Qwen3-8B variants: Base, SFT, DPO, and Instruct.
- Evaluated **142** adversarial prompts with debiased pairwise LLM judging, position-swapped comparisons, and log-prob calibration, flagging **10%** of DPO preference wins that degraded factuality.
- Diagnosed optimization-stage regressions using ECE, Brier score, AUROC, and bootstrap confidence intervals, finding SFT raised harmful compliance to **12.5%** and DPO increased sycophancy from **34%** to **45%**.

Asymmetric Relation Dynamics in Language Agents | Python, LLM Agents, LoRA, asyncio, BM25 | GitHub

- Built an async multi-agent training system (Tinker API): a frozen **Qwen3-235B** caregiver teaching LoRA-adapted **Qwen3-8B** agents via a four-module memory and salience-gated BM25 retrieval.
- Reached **100%** training success with **~25%** fewer turns (6.1 vs 8.4) over Solo/Peer baselines across **4** conditions $\times$ **3** seeds.
- Isolated scaffolding dependency as the transfer bottleneck via ablations; fully reproducible via a single-command CLI.

Decode Roofline: Kernel-Level Profiling & Fused INT4 Dequant-GEMV | CUDA C++, Nsight, PyTorch, vLLM | GitHub

- Profiled batch-1 LLM decode (**Qwen2.5-1.5B**) at the CUDA-kernel level on a consumer GPU, using Nsight to attribute **~81%** of kernel time to weight-streaming GEMVs pinned at **~95%** of the measured **~250 GB/s** memory roofline.
- Wrote a correctness-gated fused **INT4** dequant + GEMV CUDA kernel hitting **86%** of peak bandwidth (**~8.7 $\times$** fewer bytes than a two-op baseline), winning **2.38 $\times$** at batch-1 then ceding to FP16 GEMM by batch-2 as arithmetic intensity rises.

Education

Johns Hopkins University

May 2026

M.S.E. in Computer Science

GPA: 3.9/4.0

- **Coursework:** Theory of Replicable ML, Human-in-the-Loop ML, Non-Stationary Environments.

University of Mumbai

July 2024

B.Tech. in Computer Science, Honors in AI/ML

CGPA: 9.85/10

Technical Skills

Languages: Python, C++, Java, C

ML/LLM Systems: PyTorch, TensorFlow, Hugging Face, LoRA, DPO, RAG, calibration, conformal/risk control, Scikit-learn

Inference & MLOps: vLLM, CUDA C++, Nsight Compute/Systems, Docker, Kubernetes, AWS, Apache Airflow, Apache Spark

Data: SQL, PostgreSQL, MongoDB, Pandas, NumPy